

# BOE Costing Portal

## Technical Overview

The short form: the six modules, the architecture, the costing model and how it is verified.

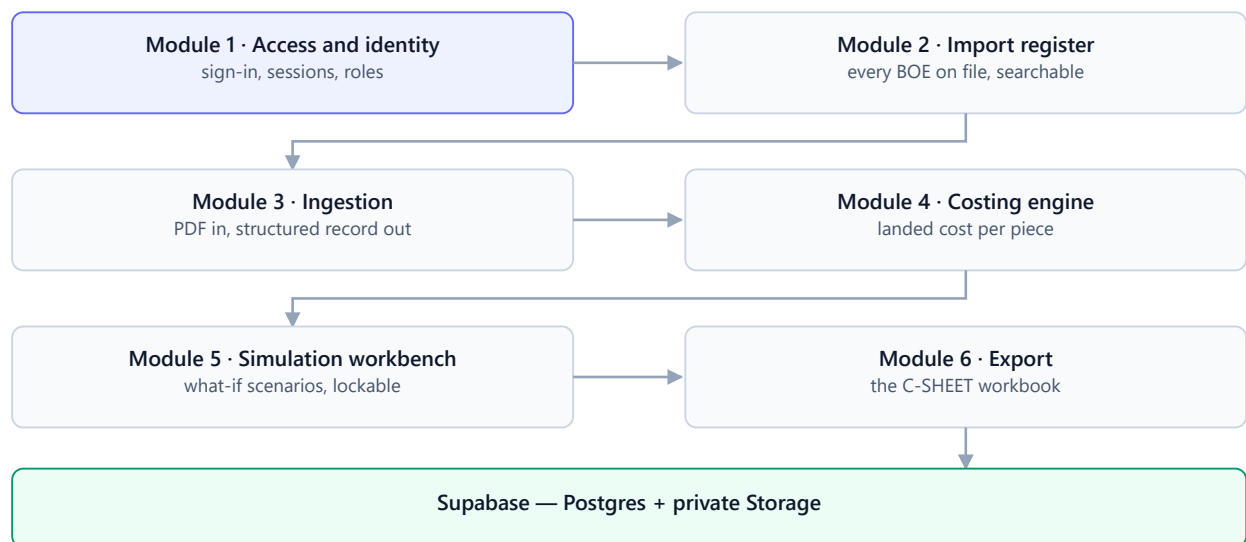
Source: docs/OVERVIEW.md

The BOE Costing Portal turns an ICEGATE Bill of Entry into a structured import record, computes the landed cost of every line item on it, and lets that costing be re-run under hypothetical inputs without disturbing the record it came from.

It answers one question, repeatedly and exactly: **what did this item cost me, landed, per piece?** The output is the C-SHEET workbook the business has always costed against — what changes is that nobody transcribes figures into it by hand, and the arithmetic behind it is pinned by tests.

This is the short form. [docs/WHITEPAPER.md](#) is the full technical document; [docs/PARSER.md](#) is the field-by-field extraction reference.

## The six modules



*Six modules, and the gate they sit behind.*

**1 · Access and identity.** The portal is entered through Supabase Auth. Every page requires a session, and the database policies grant to *authenticated* rather than to anyone, so access is decided at the data layer and not by which screen the interface chooses to render. Three credentials are deliberately unequal: the browser's anon key reads but never writes, the parser service holds the `service_role` key server-side and is the only path that can change a record, and administrative routes sit behind a token that fails closed when unset. A session also gives the system a person to attribute an action to, which is what makes scenario ownership and per-scenario locking meaningful.

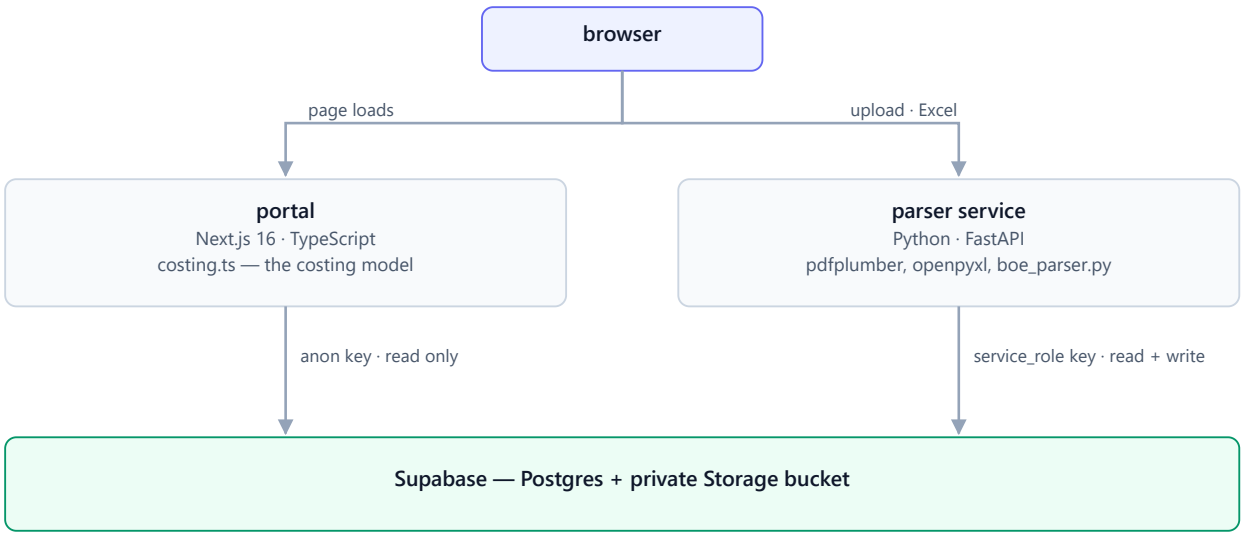
**2 · Import register.** Every Bill of Entry in one list, filtered by supplier, BOE number, date range and value range, sortable on any column. Opening a record shows its shipment facts, its costing table, its licence rows and its indexed documents. Exchange rate and freight are routinely estimated at costing time and settle weeks later, so each carries a status shown green when confirmed and red while provisional — anything never confirmed counts as provisional, because silence is not confirmation.

**3 · Ingestion.** A Bill of Entry is a fixed-layout government form, 7–30 pages, printed to PDF with no data layer. Every stored field is recovered from the printed page: labelled values by text, and the dense duty and licence grids by x-coordinate, which is the only thing that says which bare number is which. Page furniture — the watermark, the

sideways margin labels — is filtered out by character size before words are assembled, because once it fuses with a real token no text matching can separate it. Re-ingesting a record replaces its items wholesale and refreshes any figure still marked provisional, never one a person has confirmed.

- 4 • **Costing engine.** The model, below.
- 5 • **Simulation workbench.** Scenarios against one record, compared side by side.
- 6 • **Export.** The C-SHEET workbook with live formulas, for the actual record or for a scenario.

## Architecture



Two deployable services and one database. Only the parser can write.

Two deployable services and one database. The portal is TypeScript and cannot read a PDF; the extraction is positional, done with `pdfplumber` by x-coordinate, which is what lets it survive watermark bleed, OCR spelling variants and multi-invoice numbering. So one Python process exists, needed for exactly two actions — PDF upload and Excel download. Everything else is Postgres reads plus arithmetic in the browser.

They deploy as separate Vercel projects because a Next.js app and a Python function both claim `/api/*`, and inside one project Next answers first. `costing.ts` is the single implementation of the costing model: pure, with no Supabase, no React and no clock, so it is fully unit-testable. The simulation export does not redo the arithmetic — the portal posts the result it is already displaying and the parser lays it into the template, because two implementations of one model would eventually disagree.

## The costing model

Landed cost is value-proportional apportionment of a single expense pool. For each item  $i$ :

|                |       |     |                                                                          |
|----------------|-------|-----|--------------------------------------------------------------------------|
| declared INR   | $F_i$ | $=$ | $\text{unit\_price\_usd} \times \text{qty} \times \text{exchange\_rate}$ |
| value share    | $s_i$ | $=$ | $F_i / \text{SUM}(F)$                                                    |
| expense share  | $H_i$ | $=$ | $\text{expense\_pool} \times s_i$                                        |
| duty in cost   | $G_i$ | $=$ | $\text{BCD} + \text{SWS}$                                                |
| cost per piece | $I_i$ | $=$ | $(F_i + G_i + H_i) / \text{qty}$                                         |

Change any expense and every item's slice moves in proportion; change a price and the shares are recomputed before the split. The pool covers freight, insurance, clearance, misc charges, supplier freight, bank charges, own bank charges and others — all eight apportioned.

**IGST is excluded from cost.** It is a creditable input tax, so it is cash flow rather than cost; it is reported separately.

**Duty is recorded as amounts, never rates**, so effective rates are back-computed from what customs actually charged. An item's BCD may be paid in cash or met by debiting an export-incentive licence, and the form records those in different places — on a fully licence-paid Bill of Entry, page 1 shows BCD 0 while SWS is 10% of a BCD never paid in cash, and Section G is the only source of the real figure.

## Simulation

Every adjustable input on a scenario is nullable, and null means *inherit the actual*. A new scenario reproduces the actual costing exactly until something is deliberately changed, so any difference on screen is a difference the user made — which is what makes a scenario diffable against the record rather than a disconnected copy.

|            |                                                               |
|------------|---------------------------------------------------------------|
| Duty mode  | fixed at what customs charged, or floating with value         |
| Freight    | a typed total, or rate × weight, volume or containers         |
| Per item   | price and quantity overrides; FOC; duplicated rows            |
| Comparison | every scenario costed live in the browser, against the actual |

**FOC items** keep their declared value in the apportionment base — the goods still ship and are still assessed, so removing their value would shift freight onto the paid items. **Duplicated items** always derive duty from their source's rates even when duty is fixed, because locking a copy to the source's actual amounts would charge the same customs payment twice.

**A scenario may carry a password.** Once set and saved, opening it requires that password; without it the scenario is listed by name and nothing more. Enforcement is at the data layer, not in the interface: the portal reads scenario tables directly from Postgres, so a lock that only hides a panel is a curtain rather than a lock. The password is stored as a hash that is never selectable, verification returns a short-lived grant, and the row-level policy releases the scenario body only against it. The lock is on the scenario and never on the record, because the value of a scenario is being diffable against a costing that stays readable.

## Verification

Three independent checks, each catching a different class of error.

**The document reconciles against itself.** The form states its own assessable value; the parser computes one independently as `inv_value × exchange_rate + freight + insurance + misc`. The two agree to the paisa on a correct parse — BE 3168452 computes 4,204,710.42 against a stated 4,204,710.41, and BE 2898472 is exact — and a gap means an input is in the wrong unit. It is the single most useful diagnostic in the system.

**Screen and workbook agree.** The C-SHEET formulas are evaluated by hand in the test suite and checked against the engine, because both are shown to the same user from the same page.

**Invariants hold.** Switching duty mode with nothing else changed moves no number, and a new scenario reproduces the actual costing exactly. Both are tests rather than conventions.